Precision-fermented whey and egg proteins: what they cost, and where evidence would help

Memo · v2, audited (2026-09-27 01:17) · Pablo Antonio Moreno Casares · techno-economic model `pf-tea`, Humbird (2021) method, checked against NREL, Intelligen and Voutilainen designs · prepared with an AI assistant; not yet peer-reviewed

Bottom line.

- **Today's public strains are several times too expensive.** Food-grade β-lactoglobulin (BLG, whey) and ovalbumin (OVA, egg) cost $64 and $390 per kg protein at the best published titers. Whey protein traded at $9–$18 and dried egg white at $10–$21 per kg protein in 2015–2025 (2025 dollars).
- **Industry-level titers are close but not there.** At the 42 g/L private-sector average, BLG costs $21.9 with our base (price-anchored) capital estimate and $12.1 with the lighter one most published studies use. That undercuts whey in spike years like 2026 (about $34/kg protein); in 2015–2025, only 9 of 21 annual export prices were above even the lighter estimate.
- **Whey is the hardest benchmark.** It is a by-product of cheese, priced at what clears the market. Lactoferrin, a scarce milk protein, shows the other side: the same model puts PF lactoferrin at $134–$216/kg (lighter to base capital), against $200–$1,500/kg reported for the dairy product.
- **A feasibility ceiling and capital dominate.** A higher titer needs more sugar feed, and a 200 m³ fermenter that must start at least 30% full with a 50% glucose feed reaches about 74 g/L in a 72 h production phase. Both are design choices; relaxed, the secretion rate and O_2 transfer set the ceiling instead, near 125 g/L. Capital-related costs are 68% of the base-case price; sugar is 14%.
- **The physical floor is low.** With every biological parameter at its best at once, the cost is $6.1 (BLG) and $6.2 (OVA) per kg, or $3.6 and $3.6 with the lighter capital estimate. The gap between today and the floor is engineering and biology, not thermodynamics.

$/kg protein, 10 kt/yr, US	titer g/L	BLG	BLG light	OVA	OVA light
best public academic strain	13.9 / 2	$64	$32	$390	$167
base case	30	$29.4	$16.1	$29.5	$16.1
industry benchmark	42	$21.9	$12.1	$22.0	$12.1
physical limit (100 kt/yr)	119 / 118	$6.1	$3.6	$6.2	$3.6

What $20/kg requires. About 40 g/L of BLG in a 48 h production phase, or 47 g/L in 72 h, with price-anchored capital; with the lighter estimate, 21–23 g/L. $10/kg is not reachable with price-anchored capital at the base cell density and yield. With the lighter estimate it needs 45–63 g/L, near the ceiling. Beyond that, yield, cell density, production time, capital and scale must move together.

Where evidence would help most, in our judgment:

1. **Open fed-batch data.** Measured titer, productivity and sugar yield **together**, at 1–10 m³, for BLG and OVA. No public dataset has all three; the product yield in every cost estimate, ours included, is a design value.
2. **Productivity per hour of fermenter time, not titer alone.** Near the ceiling, extra titer helps only if it takes neither longer nor more feed; the workable starting volume and feed concentration at scale would move the ceiling itself.
3. **Real capital costs.** Capital is the largest cost item in every scenario, and the two estimating conventions differ by 1.8×; published costs of recent plants would settle it.
4. **Downstream data.** Diafiltration flux at scale (one patent value sizes most of the membrane area).

EU. At Dutch sugar, energy, wage and construction prices the same plant costs about $4.1/kg more. A capital charge at 4% over 20 years instead of 7.5% over 10 would cut $6.7/kg; US-level sugar $2.2. These are cost effects on one plant, not policy recommendations.

Caveats. A production cost with a 14.6%/yr capital charge, not a market price. Excludes R&D, licensing, regulatory, sales and first-of-a-kind premiums. Strain parameters rest on design values and company-reported averages (evidence grades B–C). The best public BLG titer uses an ethanol-fed production phase, priced here as glucose. Full report, audit log, calibration record and data: `pf-tea` repository.